

Yixiang Gao, Ph.D, *Electrical and Computer Engineering*
Contact: 573-825-1296 | yg5d6@mail.missouri.edu
Linkedin: <https://tinyurl.com/yixiang-linkedin> | **Google Scholar:** <https://tinyurl.com/yixiang-google scholar>
Website: <https://g1y5x3.github.io> | **GitHub:** <https://github.com/g1y5x3/>

PROFESSIONAL SUMMARY

AI/ML researcher with a Ph.D in Electrical and Computer Engineering. Started interdisciplinary collaborations with speech and language pathologists at University of Missouri - Columbia. Participated in two NIH-funded projects throughout the study. Published scientific discoveries in both engineering (EMBC, CEC) and clinical conferences (AQL). Experienced in applying AI research topics such as deep learning, computer vision, and pattern recognition into applications. Also a proud contributor of the open-source community. Eager to apply my AI/ML knowledge for solving real-world problems, while contributing to open source.

TECHNICAL SKILLS

Programming Languages : Python, C/C++/CUDA, Matlab, HTML
Libraries : pytorch, scikit-learn, tinygrad, transformers, ultralytics, mmdet, ROS, ONNX
Experienced Models : SVM, GA, MLP, CNN, RNN, transformers, ViT, DETR, YOLOs, Llama
MLOps and API Tools : Git, Docker, Kubernetes, Github Actions for CI/CD
Operating Systems : Linux(preferred), Windows
Languages : English, Chinese

EDUCATION

University of Missouri - Columbia	Ph.D in Electrical and Computer Science	2017 – Current (Expected July 2024)
University of Missouri - Columbia	BS in Electrical Engineering	2014 – 2017
University of Missouri - Columbia	BS in Computer Engineering	2014 – 2017

EXPERIENCE

Graduate Research Assistant University of Missouri - Columbia ViGIR Lab	2017 – Current (Expected July 2024)
– PhD Thesis: Confounded predictions in machine learning	2017 – Current
* Detect, quantify, and mitigate confounding factors in machine learning and deep learning models.	
– Student Investigator for NIH R01DC018026	2019 – Current
* Continuation study following the previous NIH R15 project.	
* Developed new data collection protocol and classification pipeline for both sEMG and fMRI data using pytorch and Matlab.	
* Published research on confounding factors in sEMG pattern recognition for voice pathology (CEC 2023).	
– Student Investigator for NIH R15DC015335	2017 – 2019
* Led the development of machine learning classification/clustering study using python with scikit-learn and pytorch.	
* Published pioneering research on the applications of machine learning techniques in voice pathology (AQL, EMBC, Applied Sciences).	
* Presented findings at several engineering and clinical conferences (AQL 2017/2019/2021/2023, EMBC 2018).	
Graduate Teaching Assistant University of Missouri - Columbia	2017 – 2019
– Microprocessor Engineering	
– Software Design in C and C++	
Other Projects	
– SpotMicro University of Missouri - Columbia ViGIR Lab	2022 – 2023
* An open-source quadruped robot for the MU Robotics club. Operated by a Raspberry Pi 4 and integrated with ROS.	
– Object Detection and Pose Estimation using Embedded Devices University of Missouri - Columbia ViGIR Lab	2019
* Applied YOLO on a Raspberry Pi 3 with stereo-vision cameras for object detection and pose estimation (SSCI 2019).	

PUBLICATIONS

[1] R. Farag, P. Upadhyay, **Yixiang, Gao**, et al., “COVID-19 detection from pulmonary CT scans using a novel efficientnet with attention mechanism,” in *The IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2024 (Accepted).

[2] **Yixiang, Gao**, G. N. DeSouza, M. Berardi, and M. Dietrich, “Voice sEMG classification of sentences for vocal fatigue detection using GA-SVM for confounder,” in *13th International Conference on Voice Physiology and Biomechanics (ICVPB)*, 2024 (Accepted).

- [3] **Yixiang, Gao**, M. Berardi, M. Dietrich, and G. N. DeSouza, "Feature adaptation with GA-SVM for confounding removal - an application on vocal fatigue detection using semg classification," in *The 15th Advances in Quantitative Laryngology, Voice and Speech Research (AQL)*, 2023.
- [4] **Yixiang, Gao**, M. Berardi, M. Dietrich, and G. N. DeSouza, "Removal of confounding factors using GA-SVM feature adaptation: Application on detection of vocal fatigue thru semg classification," in *IEEE 2023 Congress on Evolutionary Computation (CEC)*, 2023.
- [5] **Yixiang, Gao**, M. Dietrich, and G. N. DeSouza, "Classification of vocal fatigue using neck sEMG with leave-one-subject-out testing," in *The 14th Advances in Quantitative Laryngology, Voice and Speech Research (AQL)*, 2021.
- [6] **Yixiang, Gao**, M. Dietrich, and G. N. DeSouza, "Classification of vocal fatigue using sEMG: Data imbalance, normalization, and the role of vocal fatigue index scores," in *Applied Sciences*, vol. 11, 2021.
- [7] **Yixiang, Gao**, M. Dietrich, and G. N. DeSouza, "Explore voice production variability through neck semg clustering - challenge for accurate labeling of vocal fatigue," in *The 14th Advances in Quantitative Laryngology, Voice and Speech Research (AQL)*, 2021.
- [8] J. Demby's, **Yixiang, Gao**, and G. N. DeSouza, "A study on solving the inverse kinematics of serial robots using artificial neural network and fuzzy neural network," in *2019 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, 2019.
- [9] J. Demby's, **Yixiang, Gao**, A. Shafiekhani, and G. N. DeSouza, "Object detection and pose estimation using CNN in embedded hardware for assistive technology," in *2019 IEEE Symposium Series on Computational Intelligence (SSCI)*, 2019.
- [10] M. Dietrich, E. Tippit, A. Walker, M. Pfeiffer, **Yixiang, Gao**, and G. N. DeSouza, "Relative fundamental frequency during vocal loading and relationships with laryngeal muscular patterns," The 13th Pan-European Voice Conference, 2019.
- [11] M. Dietrich, **Yixiang, Gao**, and G. N. DeSouza, "Extralaryngeal surface emg features that distinguish between those with and without elevated scores on the vocal fatigue index," The Fall Voice Conference, 2019.
- [12] **Yixiang, Gao**, M. Dietrich, M. Pfeiffer, A. Walker, and G. N. DeSouza, "Classification of vocal gestures extracted from quasi-daily sentences to detect vocal fatigue," The 13th Advances in Quantitative Laryngology, Voice and Speech Research (AQL), 2019.
- [13] **Yixiang, Gao**, M. Dietrich, M. Pfeiffer, and G. N. DeSouza, "Classification of sEMG signals for the detection of vocal fatigue based on VFI scores," in *2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2018.
- [14] **Yixiang, Gao**, M. Pfeiffer, M. Dietrich, and G. N. DeSouza, "Classification of neck surface EMG signals for the early detection of vocal dysfunction," The 12th Advances in Quantitative Laryngology, Voice and Speech Research (AQL), 2017.

OTHER ACTIVITIES

Academic paper reviews : EMBC2019/2020/2021, CEC2023, TAI

Open-source contributions : tinygrad